No. of Functional Features Included in the Solution

Date: 17 June 2026
Team ID: XXXXXX
Project Name: **Rising Waters: A Machine Learning Approach to Flood Prediction**
Maximum Marks: 5 Marks

S.No	Feature Name	Feature Description	Module/Component	Status	Marks
1	Flood Data Collection	Collect historical flood & rainfall data	Data Collection	Done	High
2	Data Preprocessing	Clean, scale and prepare dataset	Preprocessing	Done	High
3	EDA	Visualize patterns and correlations	Analysis	Done	Medium
4	ML Model Training	Train Decision Tree, RF, KNN & XGBoost	ML Model	Done	High
5	Model Evaluation	Evaluate using Accuracy, Precision, Recall, F1	Evaluation	Done	High
6	Flood Prediction	Predict flood risk from user input	Prediction Engine	Done	High
7	Flask Web App	Interactive UI for prediction	Frontend/Backend	Done	High
8	Deployment	Load saved XGBoost model	Deployment	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features	Example
1	User Interface	2	Prediction Form, Results
2	Backend / Logic	3	Flask, Validation, XGBoost
3	Database / Storage	1	CSV Dataset, Saved Model
4	API / Integration	1	ML Integration
5	Security	1	Admin Login/Input Validation